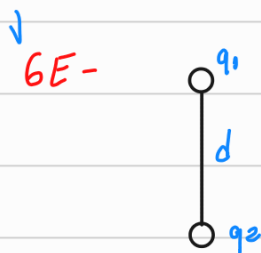


Lista 1 - Física

Trigo das Reis Gomes

Exs 6E, 8P, 9P, 10P, 13P, 14P, 15P, 17P e 19P (Cap 23)



a) $F_c = ?$ em q_1

$$q_1 = q_2 = 20 \mu C$$

$$d = 1,5 m$$

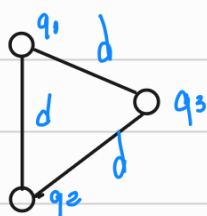
$$F_c = k \cdot \frac{q_1 \cdot q_2}{d^2}$$

$$F_c = 8,99 \cdot 10^9 \cdot \frac{20 \cdot 20 \cdot 10^{-6} \cdot 10^{-6}}{1,5^2}$$

$$F_c = 8,99 \cdot 10^9 \cdot \frac{400 \cdot 10^{-12}}{2,25}$$

$$F_c = 8,99 \cdot 10^9 \cdot 1,78 \cdot 10^{-10}$$

$$F_c = 1,6 N //$$

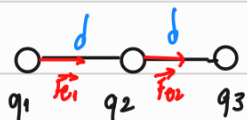


b) $F_c = ?$

$$F_c = 1,6 \cdot \cos 30^\circ + 1,6 \cdot \cos 30^\circ$$

$$F_c = 1,6 \cdot \sqrt{3} = 2,77 N //$$

8P-



$$F_{c1} = F_{c2}$$

$$k \cdot \frac{q_2 \cdot q_3}{d^2} = k \cdot \frac{q_1 \cdot q_3}{(2d)^2} \rightarrow q_2 \cdot q_3 = \frac{q_1 \cdot q_3}{4} \rightarrow q_1 = -4q_2 //$$

9P-



a) relação de q_1 e q_2 em $x = a/2$?

$$k \cdot \frac{q_1 \cdot Q}{(3/2a)^2} = k \cdot \frac{q_2 \cdot Q}{(a/2)^2}$$

$$\frac{q_1}{9/4a^2} = \frac{q_2}{1/4a^2} \rightarrow \frac{q_1}{9} = \frac{q_2}{1}$$

$$q_1 = 9q_2 //$$

b) $k \cdot \frac{q_1 \cdot Q}{(5/2a)^2} = k \cdot \frac{q_2 \cdot Q}{(a/2)^2}$

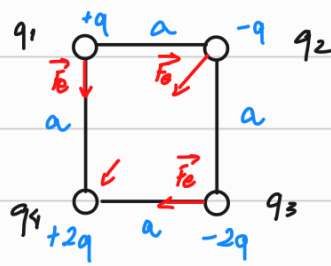
$$4q_1/25 = 4q_2$$

$$q_1 = -25q_2 //$$

10P- $F_{ex} = ?$, $F_{ey} = ?$

$$q = 1,0 \cdot 10^{-7} \text{ C}$$

$$a = 5,0 \text{ cm}$$



$$F_1 = k \cdot \frac{q_1 \cdot q_4}{a^2} = 8,99 \cdot 10^9 \cdot \frac{1 \cdot 10^{-7} \cdot 2 \cdot 10^{-7}}{(5 \cdot 10^{-2})^2} = 0,072 \text{ N}$$

$$F_2 = k \cdot \frac{q_2 \cdot q_4}{a^2} = 8,99 \cdot 10^9 \cdot \frac{1 \cdot 10^{-7} \cdot 2 \cdot 10^{-7}}{(\sqrt{2} \cdot 5 \cdot 10^{-2})^2} = 0,036 \text{ N}$$

$$F_3 = k \cdot \frac{q_3 \cdot q_4}{a^2} = 8,99 \cdot 10^9 \cdot \frac{2 \cdot 10^{-7} \cdot 2 \cdot 10^{-7}}{(5 \cdot 10^{-2})^2} = 0,144 \text{ N}$$

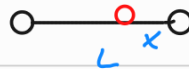
$$F_{2x} = 0,036 \cdot \cos 45^\circ = 0,025 \text{ N} = F_{2y}$$

$$F_h = 0,025 + 0,144 = 0,169 \text{ N}$$

$$F_v = 0,072 - 0,025 = 0,047 \text{ N}$$

13P- $q_1 = +1 \mu\text{C} = +1 \cdot 10^{-6} \text{ C}$
 $q_2 = -3 \mu\text{C} = -3 \cdot 10^{-6} \text{ C}$
 $d = 10 \text{ cm} = 10 \cdot 10^{-2} \text{ m} = 10^{-1} \text{ m}$

onde pode-se localizar uma 3ª carga p/ F_e seja nula?

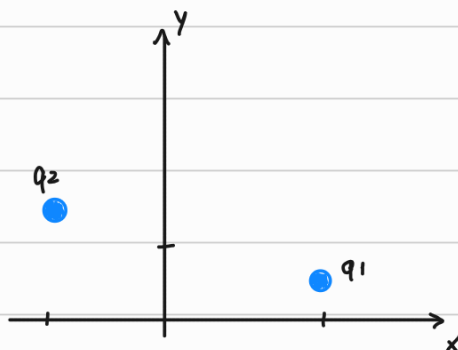


$$F_{e1} = F_{e2} \rightarrow k \cdot \frac{q_1 \cdot q_3}{x^2} = k \cdot \frac{q_2 \cdot q_3}{(L-x)^2} \rightarrow \frac{1 \cdot 10^{-6} \cdot q_3}{(10^{-1})^2} = \frac{3 \cdot 10^{-6} \cdot q_3}{(10^{-1} - x)^2}$$

$$\frac{1}{x^2} = \frac{3}{(L-x)^2} \rightarrow \sqrt{3}x = 0,1 - x \rightarrow 0,1 = x(\sqrt{3} + 1)$$

$$x = \frac{0,10}{\sqrt{3} + 1} = 0,036 \text{ m} = 3,6 \text{ cm}$$

14P- $q_1 = +3 \mu\text{C}$ $q_2 = -4 \mu\text{C}$
 $x_1 = 3,5 \text{ cm}$ $x_2 = -2 \text{ cm}$
 $y_1 = 0,5 \text{ cm}$ $y_2 = 1,5 \text{ cm}$



a) F_e em q_2 ?

$$r = \sqrt{(-2 - 3,5)^2 + (1,5 - 0,5)^2} = \sqrt{31,25} = 5,59 \text{ cm} = 0,0559 \text{ m}$$

$$F_0 = 8,99 \cdot 10^9 \cdot \frac{3 \cdot 10^{-6} \cdot 4 \cdot 10^{-6}}{0,0559^2} \rightarrow F_e = 3,45 \text{ N} //$$

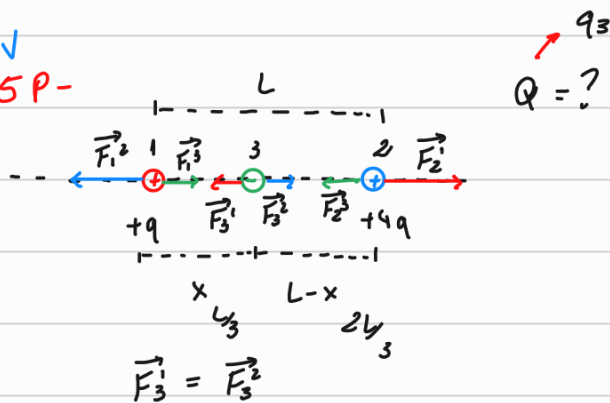
$$\text{direção: } \theta = \tan^{-1}\left(\frac{1}{-5,5}\right) = -10,3^\circ //$$

b) $F_{32} = F_{12}$

$$\cancel{k} \cdot \frac{4 \cdot 10^{-6} \cdot 4 \cdot 10^{-6}}{r^2} = \cancel{k} \cdot \frac{3 \cdot 10^{-6} \cdot 4 \cdot 10^{-6}}{0,0559^2} \rightarrow r = \sqrt{\frac{4}{3}} \cdot 0,0559^2$$

$$r = 0,0643 \text{ m} = 6,43 \text{ cm} //$$

15 P-



a carga 3 deve ser negativa para as forças totais serem 0 (sistema em equilíbrio)

$$\cancel{k} \cdot \frac{q_1 \cdot q_3}{x^2} = \cancel{k} \cdot \frac{q_2 \cdot q_3}{(L-x)^2}$$

$$Q = -\frac{4}{9} q //$$

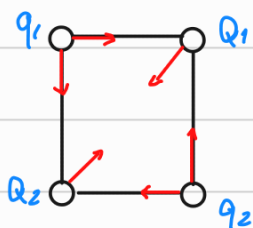
$$\frac{q \cdot Q}{x^2} = \frac{Q \cdot 4q}{(L-x)^2} \rightarrow \frac{1}{x^2} = \frac{4}{(L-x)^2} \rightarrow \frac{(L-x)^2}{x^2} = 4 \rightarrow \frac{L-x}{x} = 2 \rightarrow x = \frac{L}{3} //$$

$$F_1^4 = F_1^3$$

$$\cancel{k} \cdot \frac{q_1 \cdot q_2}{L^2} = \cancel{k} \cdot \frac{q_1 \cdot q_3}{L^2/9} \rightarrow \frac{q \cdot 4q}{L^2} = \frac{q \cdot Q}{L^2/9} \rightarrow 4q = 9Q$$

$$Q = \frac{4}{9} q //$$

17P -



a) $F_{Q1} = F_{Q2}$, $Q = ?$

$$F_{Q1} = k \cdot \frac{Q^2}{(\sqrt{2}a)^2} = k \cdot \frac{Q^2}{2a^2}$$

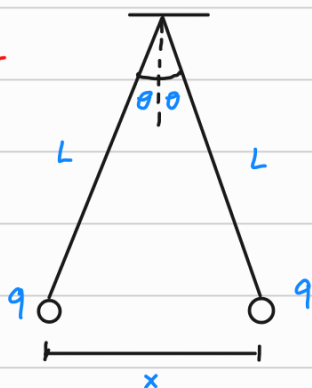
$$F_{q1} = k \cdot \frac{Q \cdot q}{a^2}$$

$$F_R = 2 \cdot k \cdot \frac{Q \cdot q}{a^2} \cdot \cos 45^\circ = 2k \cdot \frac{Qq}{a^2} \cdot \frac{\sqrt{2}}{2} = k \cdot \frac{\sqrt{2}Qq}{a^2}$$

$$k \cdot \frac{Q^2}{2a^2} = k \cdot \frac{\sqrt{2}Qq}{a^2} \rightarrow \frac{Q}{2} = \sqrt{2}q \rightarrow Q = 2\sqrt{2}q //$$

↓

19P -



a) mostrar que $x = \left(\frac{q^2 L}{2\pi \epsilon_0 m g} \right)^{1/3}$ $\tan \theta \approx \sin \theta$

$$R_x = 0 \rightarrow T \sin \theta = F \rightarrow \tan \theta = \frac{F}{mg}$$

$$R_y = 0 \rightarrow T \cos \theta = mg$$

$$F = mg \tan \theta$$

$$F = \frac{1}{4\pi \epsilon_0} \cdot \frac{q^2}{x^2} \text{ e } \sin \theta = \frac{x/2}{L} \rightarrow \sin \theta = \frac{x}{2L}$$

$$\frac{1}{4\pi \epsilon_0} \cdot \frac{q^2}{x^2} = \frac{x}{2L} mg \rightarrow q^2 L = 2x^3 \pi \epsilon_0 mg \rightarrow x = \left(\frac{q^2 L}{2\pi \epsilon_0 mg} \right)^{1/3} //$$

b) $L = 120 \text{ cm}$, $m = 10 \text{ g}$ e $x = 5 \text{ cm}$, $q = ?$

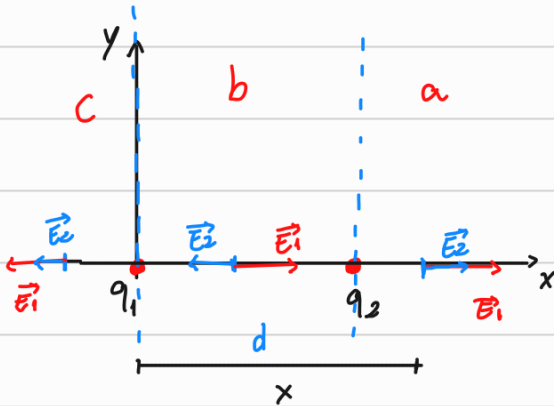
$$5 \cdot 10^{-2} = \left(\frac{q^2 \cdot 1,2}{2\pi \epsilon_0 \cdot 10 \cdot 9,8} \right)^{1/3} \rightarrow q = 2,4 \cdot 10^{-8} \text{ C} //$$

Exs 14P, 21P, 22P, 26P, 31P, 32P, 33P, 34P (Cap 24)

14P-

$$q_1 = +1 \cdot 10^{-6} \text{ C} \quad d = 10 \text{ cm} = 10 \cdot 10^{-2} \text{ m} = 10^{-1} \text{ m}$$

$$q_2 = +3 \cdot 10^{-6} \text{ C}$$



$$E(x) = ? \quad E > 0 \rightarrow$$

$$q_1 > 0 \quad E < 0 \leftarrow$$

$$q_2 > 0$$

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2} \hat{r}$$

$$a) \quad E_T = E_1 + E_2$$

$$E_T = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{x^2} + \frac{q_2}{(x-d)^2} \right)$$

$$E_T = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{x^2} + \frac{1}{(x-d)^2} \right) //$$

$$b) \quad E_T = E_1 - E_2$$

$$E_T = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{x^2} - \frac{q_2}{(d-x)^2} \right)$$

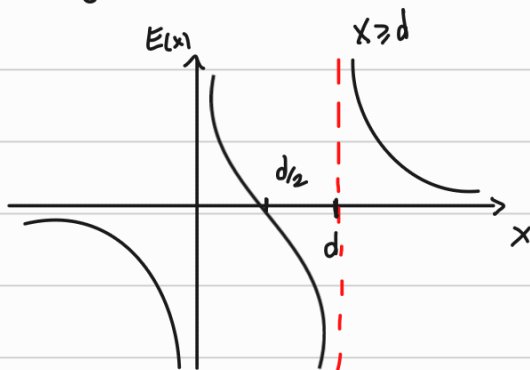
$$E_T = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{x^2} - \frac{1}{(d-x)^2} \right) //$$

$$c) \quad E_T = -E_1 - E_2$$

$$E_T = \frac{-1}{4\pi\epsilon_0} \left(\frac{q_1}{x^2} + \frac{q_2}{(x+d)^2} \right)$$

$$E_T = \frac{-q}{4\pi\epsilon_0} \left(\frac{1}{x^2} + \frac{1}{(x+d)^2} \right) //$$

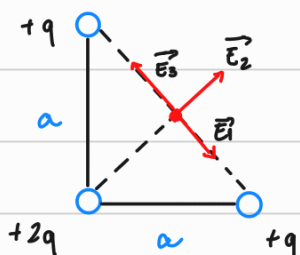
d) gráfico:



$$\frac{1}{x^2} - \frac{1}{(d-x)^2} = 0 \rightarrow \left(\frac{d-x}{x} \right)^2 = 1$$

$$d-x = x \rightarrow x = \frac{d}{2}$$

21P-

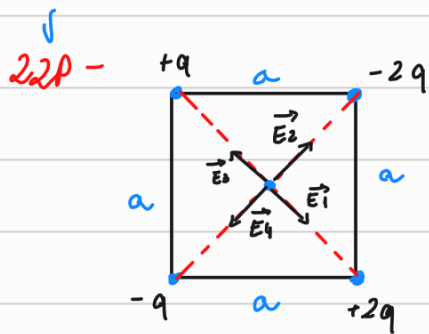


$$E_1 = E_3$$

$$\vec{E}_T = E_2 \cos \theta \hat{i} + E_2 \sin \theta \hat{j}$$

$$\vec{E}_T = \frac{q}{a^2\pi\epsilon_0} \hat{j} //$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{2q}{(\sqrt{2}/2 a)^2} \rightarrow \frac{1}{4\pi\epsilon_0} \cdot \frac{2q}{\frac{a^2}{2}} \rightarrow E_2 = \frac{4q}{a^2 4\pi\epsilon_0} = \frac{q}{a^2 \pi\epsilon_0}$$



$$q = 1 \cdot 10^{-8} \text{ C} \quad a = 5 \text{ cm} = 5 \cdot 10^{-2} \text{ m}$$

$$E_T = (E_2 - E_4) + (E_3 - E_1)$$

$$x = \frac{\sqrt{2}}{2} a$$

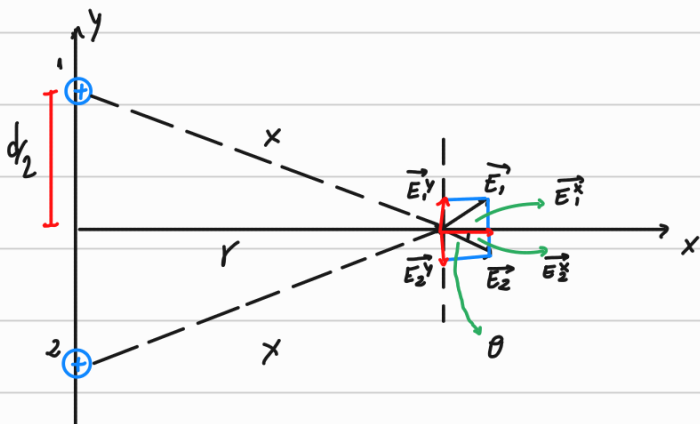
$$E_x = \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{(\sqrt{2}/2 a)^2} - \frac{q}{(\sqrt{2}/2 a)^2} \right) = \frac{1}{4\pi\epsilon_0} \cdot \frac{2q}{a^2} = 8,99 \cdot 10^9 \cdot \frac{2 \cdot 10^{-8}}{25 \cdot 10^{-4}}$$

$$E_x = 7,19 \cdot 10^4 \text{ N/C}$$

$$E_y = E_x = 7,19 \cdot 10^4 \text{ N/C}$$

$$E_T = \sqrt{2(7,19 \cdot 10^4)^2} = 1,02 \cdot 10^5 \text{ N/C} //$$

26P -



$$E_1 = E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x^2} ; \quad x^2 = r^2 + \left(\frac{d}{2}\right)^2$$

$$E_1^y = E_2^y \rightarrow E_y = 0$$

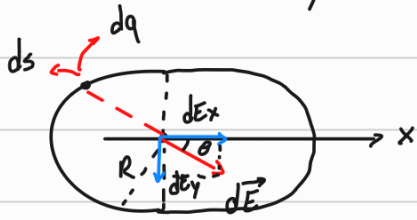
$$E_1^x = E_2^x \rightarrow E_x = 2E_1^x = 2E_2^x$$

$$E_1^x = E_1 \cos \theta ; \quad \cos \theta = \frac{r}{x}$$

$$E_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r^2 + (d/2)^2)} \cdot \frac{r}{(r^2 + (d/2)^2)^{1/2}}$$

$$\vec{E}_T = \frac{2qr}{4\pi\epsilon_0 (r^2 + (d/2)^2)^{3/2}} \hat{x}$$

31P- Por simetria, sabe-se que as duas barras produzem o mesmo campo elétrico E , $E_T = 2E$



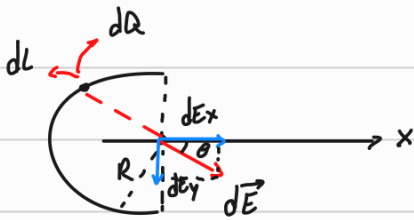
$$\lambda = \frac{dq}{ds} \rightarrow dq = \lambda ds \quad dE_x = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{R^2} \cdot \cos\theta$$

$$dq = \frac{q}{\pi R} ds, \quad ds = R d\theta$$

$$E_x = \int_{-\pi/2}^{\pi/2} \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda R d\theta}{R^2} \cos\theta = \frac{\lambda}{4\pi\epsilon_0 R} \int_{-\pi/2}^{\pi/2} \cos\theta d\theta$$

$$E_x = \frac{\lambda}{4\pi\epsilon_0 R} \cdot 2 \rightarrow E = \frac{2\lambda}{4\pi\epsilon_0 R^2} \hat{x}$$

32P-



$$\lambda_{sup} = \frac{Q}{L}, \quad \lambda_{inf} = -\frac{Q}{L}$$

$$\lambda_{sup} = \frac{Q}{\pi R}, \quad \lambda_{inf} = -\frac{Q}{\pi R}$$

$$\frac{dq}{ds} = \lambda \rightarrow dq = \lambda ds, \quad ds = R d\theta$$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{R^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda R d\theta}{R^2} = \frac{\lambda d\theta}{4\pi\epsilon_0 R}$$

$$dE_x = \frac{\lambda d\theta}{4\pi\epsilon_0 R} \cos\theta \quad e \quad dE_y = \frac{\lambda d\theta}{4\pi\epsilon_0 R} \sin\theta$$

$$E_{x, sup} = \frac{\lambda_{sup}}{4\pi\epsilon_0 R} \int_0^\pi \cos\theta d\theta = 0, \quad E_{x, inf} = 0 \quad e \quad E_{y, inf} = -\frac{2Q}{4\pi\epsilon_0 R^2}$$

$$E_{y, sup} = \frac{\lambda_{sup}}{4\pi\epsilon_0 R} \int_0^\pi \sin\theta d\theta = \frac{\lambda_{sup}}{4\pi\epsilon_0 R} (-\cos\theta) \Big|_0^\pi = \frac{2Q}{4\pi\epsilon_0 R^2}$$

$$E_x = 0 \quad e \quad E_y = 0 \rightarrow E = 0 //$$

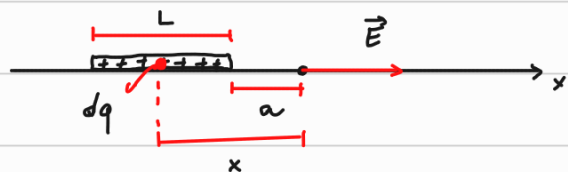
33P - $\lambda = \frac{q}{L}$ $dq = \lambda ds = \frac{q}{L} ds$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} \hat{r}$$

$$r = \sqrt{x^2 + y^2}$$

$$r =$$

34P - $E_T = \int \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{x^2}$



$$E_T = \frac{1}{4\pi\epsilon_0} \int_{x=a}^{x=a+L} \frac{dq}{x^2}$$

$$\lambda = \frac{dq}{dx} \rightarrow dq = \lambda dx$$

$$E_T = \frac{1}{4\pi\epsilon_0} \int_a^{a+L} \frac{\lambda dx}{x^2} = \frac{1}{4\pi\epsilon_0} \left. \frac{-1}{x} \right|_a^{a+L} = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{a+L} \right]$$

$$E_T = \frac{1}{4\pi\epsilon_0} \cdot \frac{a+L - a}{a(a+L)} \rightarrow \vec{E}_T = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda L}{a(L+a)} \hat{x}$$